🔍 Exploratory / Analytical SQL Questions

1. Which store has sold the **highest number of units** overall?
2. Which **product category** generates the **highest total revenue**?
3. List the **top 5 products** with the **highest total warranty claims**.
4. Find the **average selling price** of each product **per store**.
5. What is the **store-wise contribution (%)** to the overall sales quantity?
6. List all products that were **launched before 2020** and are **still generating sales**.
7. Identify the **month** with the **highest number of sales** across all years.
8. List the **top 3 cities** with the highest **product diversity sold** (distinct product count).
9. Find the **product(s)** with the **highest warranty-to-sales ratio**.
10. Calculate the **average warranty claim time** (in days) between `sale_date` and `claim_date`.

📊 Time Series Analysis

11. Show the **year-over-year growth** in total sales revenue.
12. List the **monthly average revenue** generated from each product in 2024.
13. Identify any product that had **zero sales for three consecutive months**.
14. Find the **earliest and latest sale** for each product and the time span between them.
15. Which stores saw a **decline in sales volume** compared to the previous year?

🧩 Complex Joins & Multi-table Insights

16. List the **stores** that **sold products from all categories**.

17. Which store had the **highest number of different warranty repair statuses**?

18. List the products which were **sold in all countries**.

19. For each country, find the **most frequently sold product category**.

20. Identify stores where **every sale** led to a **warranty claim**.

🧠 Logic and Pattern-Based

21. List products that were **sold within 7 days** of their launch date.

22. Identify stores that have **never sold a product with price > $1000**.

23. Find all **products that have more than 3 warranty claims**, and **all occurred within 30 days of sale**.

24. List **sales** where the **quantity is more than 3 times the product's average quantity sold**.

25. Identify **warranty claims** that were made on **public holidays or weekends** (use day of week).

💡 Revenue & Profitability-Oriented

26. Calculate the **total revenue lost** due to products that had a **warranty claim** with repair status = 'Defective'.

27. Find the **average revenue per product per year**.

28. Which product had the **highest average sale value** per transaction?

29. Show the **top-selling product in terms of revenue for each country**.

30. Which category has the **highest price variance** among its products?

🧮 Advanced Metrics & Business KPIs

31. Calculate the **sales conversion rate** = (number of sales with warranty claim / total sales) per product.

32. Show **store-product pairs** with consistent sales every month in 2023.

33. Compute the **lag in days** between each sale and its corresponding warranty claim (if any), and **rank the top 5 fastest claims**.

34. Identify **products that haven’t been sold in the past year** but had warranty claims.

35. Calculate the **average time to warranty claim** per category and rank them.